

Introduction to the Web and the Internet

ITELEC 4100 - Lecture



TOPICS

- History of the Internet
- Networks and Networking
- The Internet
- The World Wide Web
- Uses of the Internet
- Internet Protocols
- Web Hosting

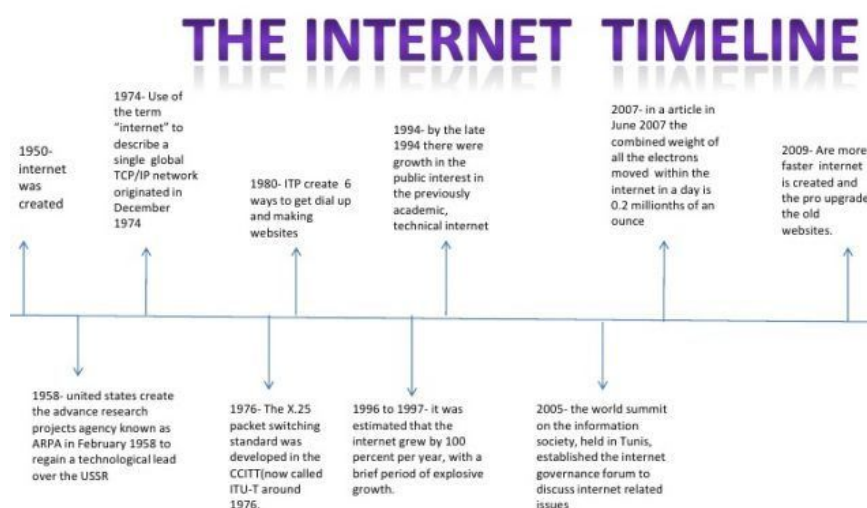


History of the Internet

The history of the Internet started way back 1950s, when the world was in the Cold War. Researchers strive to create a system for communication, from military projects to the global web as we know it today.



History of the Internet



History of the Internet

Key Milestones in Internet History

- **1969:** Launch of ARPANET, the first operational packet-switching network.
- **1971:** The first email is sent by Ray Tomlinson.
- **1983:** Adoption of TCP/IP as the standard protocol for ARPANET.
- **1989-1990:** Tim Berners-Lee invents the World Wide Web.
- **1993:** The first graphical web browser, Mosaic, is released.
- **1998:** Google is founded, revolutionizing search engines.
- **2007:** The iPhone introduces widespread mobile Internet access.
- **2010s:** Rise of cloud computing, social media, and streaming services.



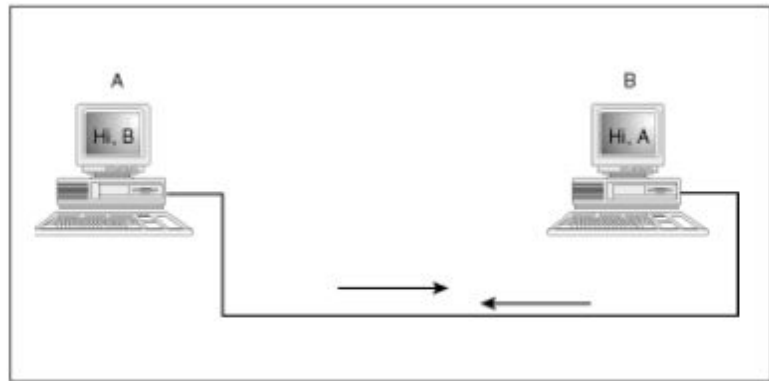
Networks and Networking

- Network – a group of interconnected systems sharing services and interacting by means of a shared communications link.
- In summary, all networks must have the following:
 - Something to share (data)
 - A physical pathway (transmission medium)
 - Rules of communications (protocols)



Networks and Networking

- At its simplest, a **computer network** is two or more computers/devices sharing information across a common transmission medium.



A simple computer network

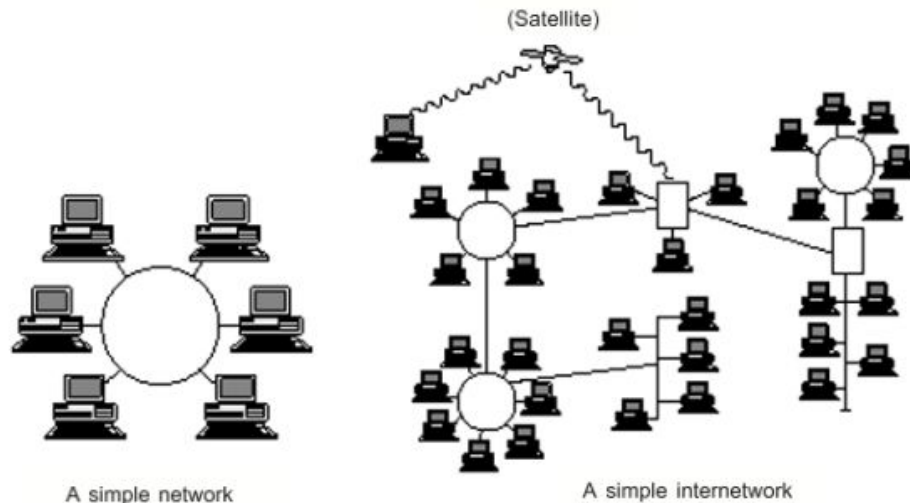


Networks and Networking

- Internetwork – two or more devices (computers) are connected so that they can exchange messages and share information.
- internet (lowercase i) is short for Internetwork.
- Any computer on an Internetwork can talk to any other, but the message may have to travel.



Networks and Networking



The Internet

- Internet (uppercase I) is the world's largest Internetwork.
- IP address is a unique identification, each computer in the Internet has an assigned IP address.
 - InterNIC an organization that gives IP address to various Internet Service Providers (ISP's).
 - A survey says that there are 4,294,967,296 possible IP addresses on the Internet.
- Domain name is an "English" version of an IP address.

The Internet

- Uses a client-server relationship, which means that not all computers connected to the Internet are the same.
- Servers are special types of computers that are capable of delivering information and data.
- Clients, are computers that may only receive data.



The World Wide Web

- (WWW) or simply the Web, is the fastest growing part of the Internet.
- An information system on the Internet which allows documents to be connected to other documents by hypertext links.
- Note that, contrary to what many people believe, **the Web and the Internet are not the same.**



Basic elements of the Web

- Hypertext Markup Language (HTML) is the foundation for all Web pages
- Webmaster – the individual responsible for maintaining and updating the content of a WWW document
- Domain name – the name given to any computer registered on the WWW as official provider of info and files
- Uniform Resource Locator – serves as identification for all WWW documents.



URL TYPES

Service	Server type and Domain Name
Web	http://www.server.name
Gopher	gopher://gopher.server.name
FTP	ftp://ftp.server.name
Telnet	telnet://server.name
E-mail	mailto:address@server.name



Related terms

- Web site
- Home page
- Link
- Anchor
- Image map
- Frame
- Table



Uses of the Internet

- E-mail
- Chatting
- Getting files



Uses of the Internet

- E-mail still makes up a majority of Internet traffic today because almost everyone with Internet access has an e-mail account.
- Three elements of e-mail address:
 - User ID
 - @
 - Domain
- E-mail message contains two basic parts: header and body



Uses of the Internet

- Different headers when sending an e-mail
- **To:**
- **From:**
- **Subject:**
- **CC:**
- **BCC:**
- **Attachments:**



Uses of the Internet

- Chatting – at some point, you may want to remove the time delay between sending and receiving e-mail by conversing directly with another person or a group of people.
- Internet Relay Chat (IRC), a multiuser version of a program called Talk.
- IRC was developed in the late 1980s by Jarkko Oikarinen.
- Consists of a network of chat servers located all over the globe.



Uses of the Internet

- Getting files – **File Transfer Protocol (FTP)** is one of the most popular methods of obtaining software and documents on the Internet.
- FTP is basically a set of rules for transferring files in a fast, efficient, and reliable way.
- It was one of the first Internet services developed to enable users to transfer files from one place to another.
- FTP sites are the servers that keep files available for FTP



Navigating the Web

- Basic parts of a Web is URL or address of an Internet file.
- Usually it consists of four parts: protocol, server (or domain), path, and filename.
- Sometimes there's no path or filename.
- Example
 - <http://www.microsoft.com/windows/default.asp>
- Note that the files stored in the Internet are basically stored the same way as files in your computer.



Navigating the Web

- According to Alex Chris of Reliabelsoft Digital Marketing Agency (March 28, 2023).
 - The 10 best search engines in 2023, ranked by popularity.
1. Google (92.57%)
 2. Microsoft Bing
 3. Yahoo
 4. Baidu
 5. Yandex
 6. DuckDuckGo
 7. Ask.com
 8. Ecosia
 9. Aol.com
 10. Internet Archive



Navigating the Web

- Search engine is a web service that indexes, organizes, and often rates and reviews Websites.
- Helps you find the one needle you're looking for in the Internet haystack.
- Different search engines work in different ways:
 - Some rely on people to maintain a catalog of Web sites or pages
 - Some use software to identify key information on sites across the Internet
 - Some combine both types of service



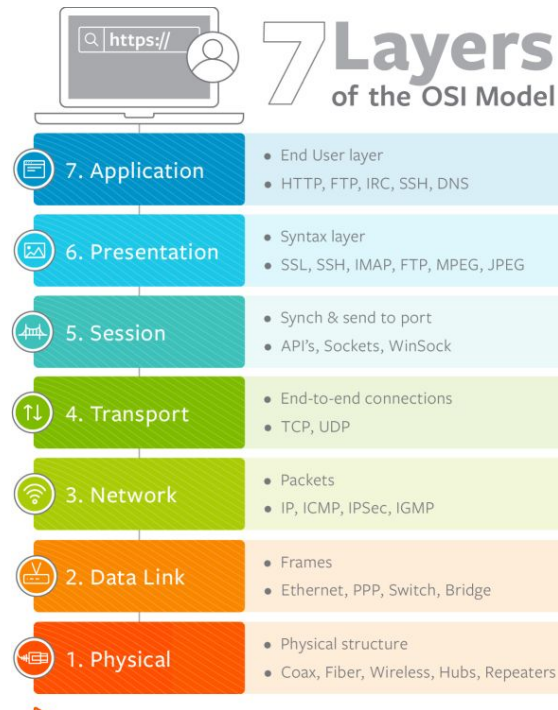
Internet Protocols

- Protocols are defined as ways in which computers communicate with each other.
- They determine how data is sent, how data is received, what must be done to ensure that the data received is correct, what kind of data is sent, and so on.
- The developers of the Internet chose certain protocols to be used for the Internet



Internet Protocols/IP

- The International Standards Organization (OSI) developed a reference model that identifies various levels in network communication.



Internet Protocols

- Transmission Control Protocol – Internet Protocol (TCP/IP)
- Was adopted as the standard by the developers of the Internet.
- The Internet is in fact a worldwide connection of interconnected networks that uses TCP/IP to connect various physical networks in to a single logical network.
- In short, TCP/IP is required for Internet connection.

Internet Protocols

- Hypertext Transfer Protocol (HTTP)
- is the World Wide Web application-layer protocol.
- Implemented in two programs:
 - A client program
 - A server program
- HTTP defines how Web clients (Internet browsers) request pages from Web servers and how Web servers transfer Web pages to Web clients.



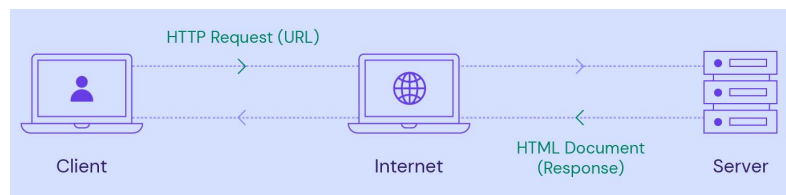
Internet Protocols

- A Web page, also called a document, consists of objects which may be an HTML file, images, applets and so on.
- A Web page is usually composed of an HTML file and several referenced objects.



Internet Protocols

- HTTP basically works in the following way:
 1. HTTP client sends a request to the server for a particular Web page
 2. TCP/IP connects to the appropriate host.
 3. HTTP server sends back the HTML file to the Web client.
 4. If HTML file references several object (images, applets, audio clips), HTTP client request for these objects



How HTTP works



Internet Addressing

- Uniform Resource Locator (URL) refers to the address of a resource on the Web.
- A resource is anything available on the Web like an HTML page, a graphic file or a CGI script.
- Uniform Resource Name (URN) names a specific resource on the Web.
- Resources can be identified using either URN or URL.
- Both URL's and URN's are components of the more general class called Uniform Resource Identifiers (URI)



Internet Addressing

- The following is a list of all protocols currently available for use with URL's:
 - HTTP
 - FTP
 - File
 - Mailto invokes SMTP
 - Telnet
 - NNT
- The choice of the protocol used depends on the type of communication you want done and task you want accomplished.



Internet Addressing

- After the protocol comes the **resource description**.
- A colon (“:”) separates the resource description from the protocol.
- The resource description may consist of any of the following elements:
 - Site address (Host)
 - Port number
 - Filepath and filename
 - Fragment identifier
 - Parameter



Internet Addressing

- **Site address** the host element identifies the site address of the computer which is the source of the file being displayed.
- Consists of two parts, the machine name and the domain name.
 - `http://165.220.21.45`
 - `http://www.example.com`
- The second URL above is pointing to a computer named `www` on the `example.com` domain.



Internet Addressing

- **Port number** follows the site address but is separated from it using a colon.
- Indicating a port number is optional.
- If you omit this element in a URL, the default port number will automatically point to 80.
- All HTTP requests are normally answered on port 80 (HTTPS requests in port 443)



Internet Addressing

- **Filepath and filename.**
- What is not known to the user is the fact that whenever you access a resource on the Web or the Internet, you are actually pointing to a file.
- When you type the address
 - <https://www.yahoo.com>
- You are actually accessing the resource:
 - <https://www.yahoo.com/index.html>



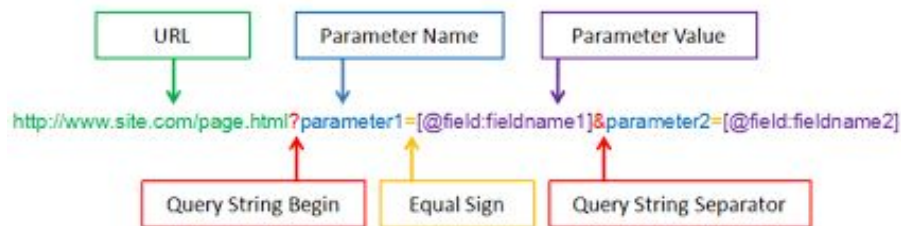
Internet Addressing

- **Fragment identifier**
- This feature of URL's allows you to actually point to a specific location (a section, paragraph, or sentence) within a file to view.
- Example:
 - <http://www.homehealthcare.com/childhood/rubella.html#tretment>
- The URL scrolls the page right down to the section of discussing treatment for common measles.



Internet Addressing

- **Parameter** one last component that may be included in a URL
- Parameters are arguments or data variables passed on to the next resource being retrieved.
- Example
 - `http://example.com?productid=1234`



Internet Addressing



Web Hosting

- **Web Hosting** provides a place to store a web content online. The images, video, and text that comprise a website. Also the **codes/program** and most of the time web hosting also provide the language compiler/interpreter.



Web Hosting Startup

- **Register a domain:** buy from domain name registrar.
 - **Ex. Google, GoDaddy, BlueHost, etc..**
- **Nameservers (DNS):** the middle man that points your website toward your domain IP Address.
 - most of the domain name registrar/company provider will also provide this together with the domain.
- **Web Hosting:** the actual file that makes up the website reside in a server and commonly maintained by the web hosting provider/company



Common Type of Web Hosting

- **Shared Hosting** - A common and economical option, shared hosting services manage multiple clients from the same server system. It's a great option when your needs and budget are limited, but it comes with a trade-off. There's more of an opportunity for disruption because the server's resources are distributed between multiple users.
- **Virtual Private Servers – (vps)** similar to shared hosting but the files are stored in a dedicated space with dedicated resources.



Common Type of Web Hosting

- **Dedicated Hosting** - This provides a dedicated server, which cuts the security and performance risks associated with shared hosting. You can also gain substantial functionality, though this increases the need for regular management and maintenance, and will increase costs.
- **Managed Hosting** – have access to a dedicated server, but you don't have the same administrative options. Managed hosting leaves security and maintenance responsibilities with your provider, leaving you to manage content via FTP (File Transfer Protocol), the rules that networked computers use when they “talk” to each other.



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